

BIOINFORMATICS

Lecture 1 — Introduction to Bioinformatics

Foundations of Bioinformatics & Biological Data

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**DNA • DATA • COMPUTING •
AI**

Visual placeholder / suggested image

Learning Objectives

By the end of this lecture, students should be able to:

Define bioinformatics and explain its purpose.

Differentiate computational biology from bioinformatics.

Explain why biological data has unusual scale, structure, noise and uncertainty.

Describe how Data Science supports biological discovery.

Identify major applications of bioinformatics.

Navigate NCBI, UniProt and Ensembl at a basic level.

Learning pathway
Concept → Data → Tools
→ Analysis →
Interpretation

Visual placeholder / suggested
image

What Is Bioinformatics?

Bookish definition + classroom explanation

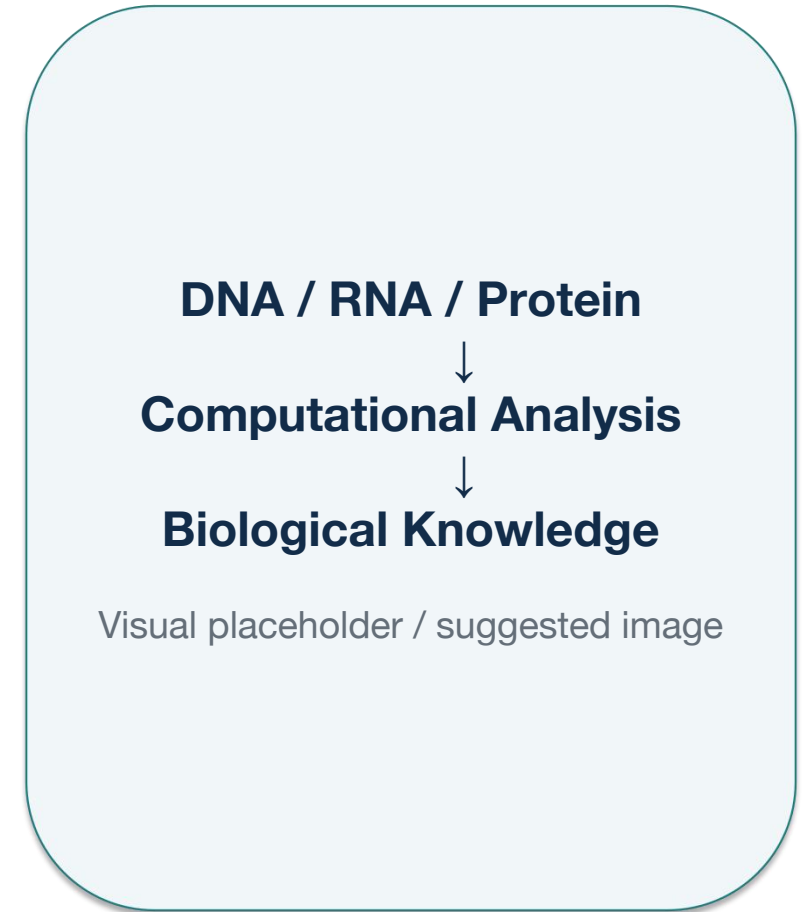
Definition:

Bioinformatics is the application of computational methods, algorithms, databases and statistics to store, manage, analyze and interpret biological data.

In simple words:

Bioinformatics uses computers to answer biological questions from data such as DNA sequences, RNA expression, protein sequences and genomes.

It combines biology + computer science + statistics + data science.



Why Do We Need Bioinformatics?

Modern biology generates data at a scale humans cannot analyze manually.

Sequencing technologies can produce millions to billions of DNA/RNA reads.

Biological databases contain enormous collections of sequences, annotations and experimental results.

Biological patterns may be subtle and distributed across thousands of variables.

Computational methods make it possible to search, compare, classify, model and visualize these patterns.

Bioinformatics turns raw biological measurements into interpretable evidence.

DATA SCALE

**Small experiment →
Millions of measurements**

**Human inspection ×
Computational analysis ✓**

Visual placeholder / suggested image

Computational Biology vs Bioinformatics

Related fields, but not identical

Bioinformatics:

Focuses strongly on biological data management, algorithms, databases and analysis pipelines.

Examples: sequence alignment, genome annotation, variant analysis, biological databases.

Computational Biology:

Uses mathematical and computational models to understand biological mechanisms and systems.

Examples: protein structure modeling, molecular simulation, systems biology, evolutionary modeling.

Overlap:

Both use computation to understand biological systems; the boundary is often practical rather than strict.

BIOINFORMATICS
Data & information



**COMPUTATIONAL
BIOLOGY**
Models & mechanisms

Visual placeholder / suggested image

Bioinformatics as a Data Science Problem

Biological question → define what we want to know.

Data acquisition → obtain sequences, expression data, variants, clinical data or other measurements.

Preprocessing → quality control, filtering, normalization and formatting.

Feature extraction → convert biological information into useful numerical or categorical features.

Modeling → statistics, machine learning, deep learning or biological algorithms.

Interpretation → translate computational output into biological meaning.

DATA SCIENCE PIPELINE

**Question → Data →
Processing
→ Features → Model →
Insight**

Visual placeholder / suggested image

Why Biological Data Is Different

Biological data has special properties that affect analysis.

High dimensionality:

Thousands or millions of genes, variants or molecular measurements.

Hierarchical structure:

DNA → genes → transcripts → proteins → pathways → phenotypes.

Noise & uncertainty:

Sequencing errors, batch effects, measurement variation and incomplete annotation.

Context dependence:

The same gene or variant can behave differently across tissues, conditions or populations.

Rapid growth:

New experiments continuously add data to public repositories.

BIOLOGICAL DATA
High-dimensional
Noisy
Context-dependent
Evolving

Visual placeholder / suggested image

Common Types of Biological Data

Genomic data — DNA sequence and genetic variation.

Transcriptomic data — RNA abundance and gene expression.

Proteomic data — proteins, abundance and modifications.

Epigenomic data — DNA methylation, chromatin accessibility and regulatory marks.

Metabolomic data — small molecules and metabolic states.

Single-cell data — measurements at individual-cell resolution.

Spatial omics — molecular measurements linked to tissue location.

Clinical/phenotypic data — patient traits, diagnoses and outcomes.

MULTI-OMICS
DNA → RNA → Protein →
Metabolites
+ phenotype + spatial
context

Visual placeholder / suggested
image

Role of Data Science in Biology

Data collection — organize large biological datasets and metadata.

Data cleaning — detect missing, duplicated, low-quality or inconsistent observations.

Exploratory analysis — identify patterns, distributions and relationships.

Statistics — quantify uncertainty and test biological hypotheses.

Machine learning — classify samples, predict outcomes and discover biomarkers.

Deep learning — learn complex representations from sequences, images and molecular data.

Visualization — communicate patterns and biological findings.

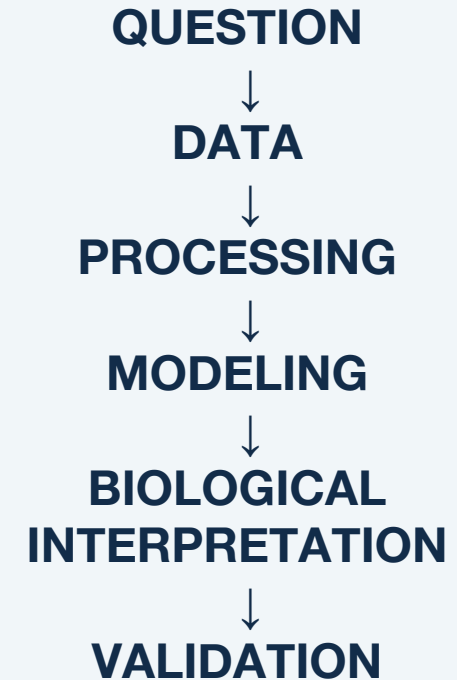
Reproducibility — preserve code, environments, parameters and provenance.

DATA SCIENCE TOOLKIT
Python • R • SQL • ML • DL
Statistics • Visualization •
Workflows

Visual placeholder / suggested
image

The Bioinformatics Data Science Pipeline

1. Biological Question — What do we want to discover?
2. Data — What measurements can answer the question?
3. Processing — How do we make the data reliable?
4. Modeling — Which statistical/ML/computational method is appropriate?
5. Biological Interpretation — What does the result mean?
6. Validation — Can the finding be reproduced or independently supported?



Visual placeholder / suggested image

Healthcare

Application of bioinformatics

Definition / role:

Use genomic and molecular data to support disease research, diagnosis, biomarker discovery and clinical decision support.

Typical applications:

Genomics • diagnostics • biomarkers • clinical research

Data Science contribution:

Integrate heterogeneous data, build predictive models, quantify uncertainty and prioritize candidates.

BIOINFORMATICS IN HEALTHCARE

**Data → Analysis →
Decision**

Visual placeholder / suggested image

Drug Discovery

Application of bioinformatics

Definition / role:

Use biological and chemical data to identify targets, understand mechanisms and prioritize candidate compounds.

Typical applications:

Target identification • virtual screening • structure prediction

Data Science contribution:

Integrate heterogeneous data, build predictive models, quantify uncertainty and prioritize candidates.

BIOINFORMATICS IN DRUG DISCOVERY

**Data → Analysis →
Decision**

Visual placeholder / suggested image

Agriculture

Application of bioinformatics

Definition / role:

Use genomics and phenotypic data to study crops, livestock, pathogens and traits important for yield and resilience.

Typical applications:

Plant genomics • breeding • pathogen surveillance

Data Science contribution:

Integrate heterogeneous data, build predictive models, quantify uncertainty and prioritize candidates.

BIOINFORMATICS IN AGRICULTURE

**Data → Analysis →
Decision**

Visual placeholder / suggested image

Infectious Diseases

Application of bioinformatics

Definition / role:

Compare pathogen genomes, track variants and investigate transmission, resistance and evolution.

Typical applications:

Pathogen genomics • outbreak analysis • AMR

Data Science contribution:

Integrate heterogeneous data, build predictive models, quantify uncertainty and prioritize candidates.

BIOINFORMATICS IN INFECTIOUS DISEASES

**Data → Analysis →
Decision**

Visual placeholder / suggested image

Precision Medicine

Application of bioinformatics

Definition / role:

Combine molecular and clinical information to understand why treatments work differently among individuals.

Typical applications:

Patient stratification • pharmacogenomics • biomarkers

Data Science contribution:

Integrate heterogeneous data, build predictive models, quantify uncertainty and prioritize candidates.

BIOINFORMATICS IN PRECISION MEDICINE

**Data → Analysis →
Decision**

Visual placeholder / suggested image

Biotechnology

Application of bioinformatics

Definition / role:

Support engineering and optimization of biological systems, proteins, enzymes and production organisms.

Typical applications:

Protein engineering • synthetic biology • bioprocessing

Data Science contribution:

Integrate heterogeneous data, build predictive models, quantify uncertainty and prioritize candidates.

BIOINFORMATICS IN BIOTECHNOLOGY

**Data → Analysis →
Decision**

Visual placeholder / suggested image

Case Study: From DNA Data to a Health Insight

Question: Which genetic variants may be associated with a disease?

Collect: sequencing reads + reference genome + sample metadata.

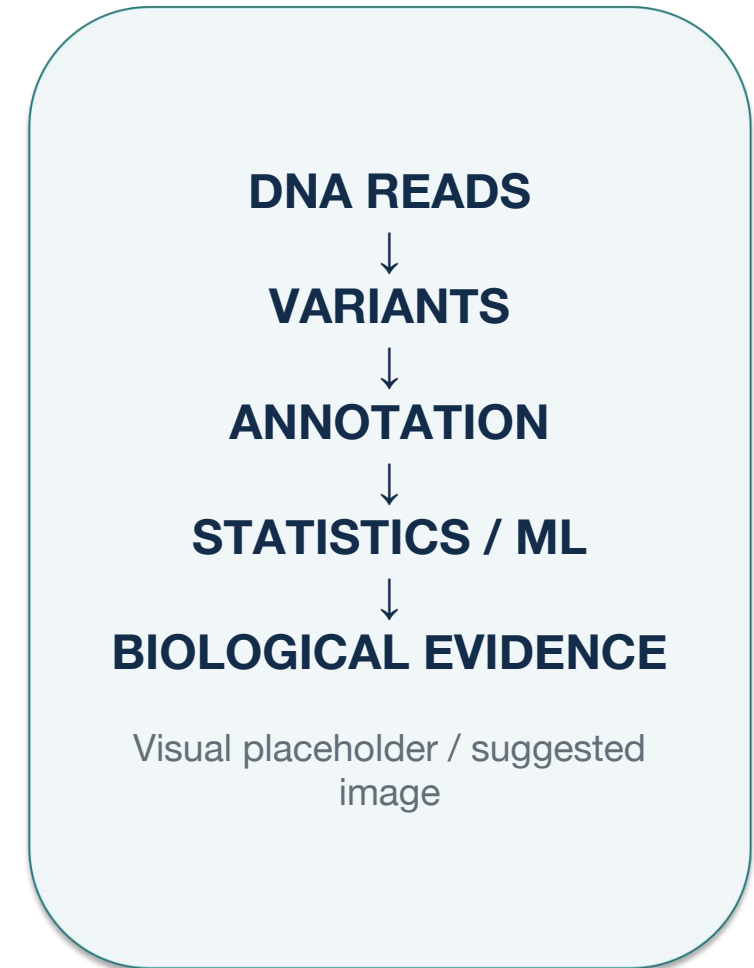
Process: quality control → alignment → variant calling.

Analyze: filter and annotate variants; compare cases and controls.

Model: statistical association or machine-learning prediction.

Interpret: identify biologically plausible variants and supporting evidence.

Important: computational association is not automatically proof of causation.



Bioinformatics Tools & Resources

NCBI:

A major biomedical information resource providing databases, search/retrieval systems, BLAST and APIs/tools.

UniProt:

A comprehensive resource for protein sequence and functional information, including reviewed and automatically annotated records.

Ensembl:

Provides integrated genome annotation and tools for exploring genomes across many species.

Key principle:

Use authoritative databases, record accession IDs, versions and search criteria, and document how data were obtained.

NCBI
GENES • GENOMES •
LITERATURE

UniProt
PROTEINS • FUNCTION

Ensembl
GENOMES • ANNOTATION

Visual placeholder / suggested
image

Practical 1 — Explore NCBI

Hands-on activity

Open NCBI and locate the Gene database.

Search for a familiar human gene, such as TP53.

Inspect gene name, organism, genomic location and links to related resources.

Open the nucleotide/protein records associated with the gene.

Identify the accession identifier and understand why it matters.

Discuss: What information is biological data and what information is metadata?

CLASS ACTIVITY

**Search → Inspect →
Record**

**Students should note:
Gene ID • accession •
organism
sequence • annotation**

Visual placeholder / suggested image

Practical 2 — Explore UniProt

Protein information and functional annotation

Search UniProtKB for a protein/gene of interest.

Identify the protein name and organism.

Inspect the amino-acid sequence.

Explore function, subcellular location, domains/features and evidence.

Compare reviewed (Swiss-Prot) and unreviewed (TrEMBL) records.

Discuss why annotation evidence and review status matter.

UNIPROTKB
Sequence
Function
Domains
Evidence
Cross-references

Visual placeholder / suggested
image

Practical 3 — Explore Ensembl

Genome annotation and gene context

Search for a human gene.

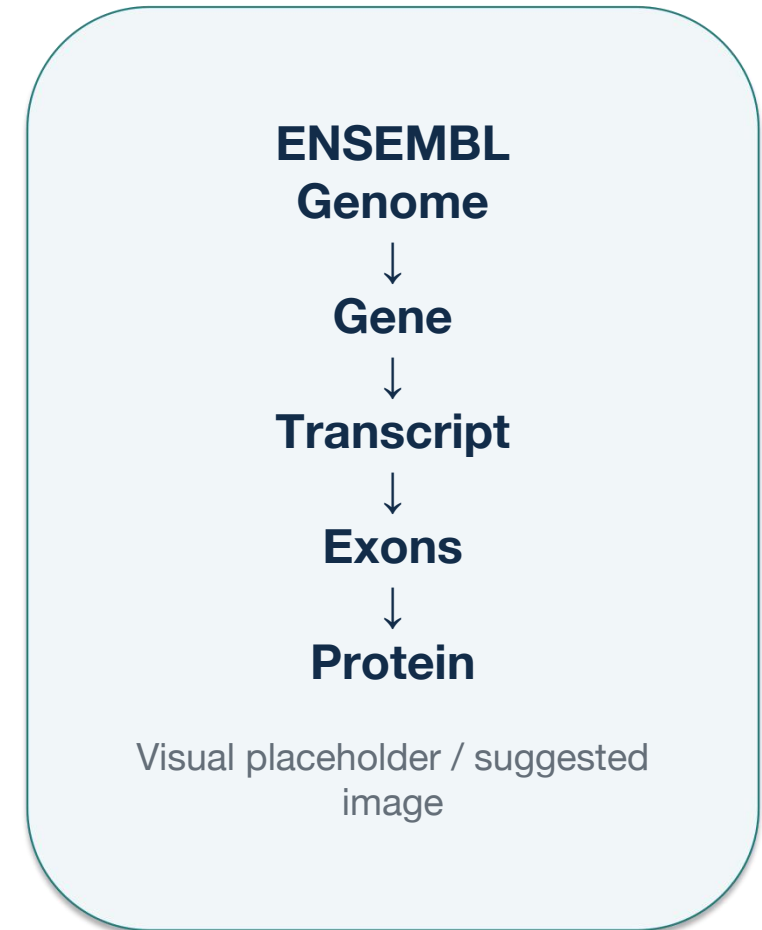
Inspect the gene location and genomic region.

Explore transcripts and alternative transcripts.

Inspect exons, transcripts and protein products.

Review comparative genomics information where available.

Discuss: How does a genome browser connect sequence with biological annotation?



Important Data Science Concepts for Bioinformatics

Feature — measurable property used for analysis or prediction.

Label — known outcome used in supervised learning.

Metadata — information describing samples, experiments and conditions.

Normalization — transformation used to make measurements comparable.

Batch effect — systematic technical variation unrelated to the biological question.

Overfitting — a model learns noise or idiosyncrasies instead of general biological patterns.

Reproducibility — another researcher can repeat the computational analysis and obtain consistent results.

**BIOLOGICAL DATA
+ METADATA
+ STATISTICS
+ COMPUTATION
= REPRODUCIBLE
EVIDENCE**

Visual placeholder / suggested
image

Common Beginner Mistakes

Treating database annotations as unquestionable truth.
Using a dataset without checking metadata and experimental design.
Ignoring sequencing or measurement quality.
Confusing correlation/association with causation.
Training ML models without proper train/test separation.
Reporting p-values without considering effect size or multiple testing.
Failing to record software versions, parameters and data accessions.
Using AI-generated biological claims without independent verification.

GOOD BIOINFORMATICS
Question → Evidence →
Validation

Not:
Tool → Output →
Assumption

Visual placeholder / suggested
image

Mini Classroom Activity

Think like a data scientist

Scenario: A researcher gives you a DNA sequence from an unknown organism.

Question 1: What database would you search first?

Question 2: What information would you need before analyzing it?

Question 3: Which computational method could identify similar sequences?

Question 4: How would you decide whether a match is convincing?

Question 5: What additional evidence would you need before making a biological claim?

UNKNOWN SEQUENCE



DATABASE SEARCH



SIMILARITY



EVIDENCE



INTERPRETATION

Visual placeholder / suggested
image

Key Takeaways

Bioinformatics applies computation and statistics to biological data.

Computational biology emphasizes computational modeling of biological systems; bioinformatics emphasizes biological data and information analysis, with substantial overlap.

Biological data is large, high-dimensional, noisy, structured and context-dependent.

Data Science provides the methods for cleaning, exploring, modeling and interpreting biological data.

Bioinformatics supports healthcare, drug discovery, agriculture, infectious disease research, precision medicine and biotechnology.

NCBI, UniProt and Ensembl are foundational resources for practical exploration.

Good bioinformatics requires biological reasoning, computational rigor and reproducibility.

BIOLOGY
+
DATA SCIENCE
+
COMPUTING
+
STATISTICS
=
BIOINFORMATICS

Visual placeholder / suggested
image

Discussion Questions

Check understanding

1. What makes biological data different from ordinary tabular data?
2. What is the difference between bioinformatics and computational biology?
3. Why are databases essential to modern biology?
4. How can machine learning help analyze biological data?
5. Why is metadata important?
6. Why should a bioinformatics result be biologically validated?
7. When would you use NCBI, UniProt and Ensembl?

DISCUSS
PAIR → SHARE →
JUSTIFY

Visual placeholder / suggested
image

Suggested Homework / Assignment

Build your first biological data investigation

Choose one human gene or protein.

Use NCBI to obtain basic gene/sequence information.

Use UniProt to investigate the associated protein.

Use Ensembl to inspect genomic location and transcripts.

Create a one-page comparison of what each resource provides.

Include accession/identifier information and screenshots.

Write 3–5 sentences explaining how the resources complement each other.

DELIVERABLE
1-page report
+ screenshots
+ identifiers
+ interpretation

Visual placeholder / suggested
image

Recommended Visuals for the Instructor

Use real screenshots during classroom delivery

NCBI homepage and Gene database search screen.

Example NCBI gene record showing identifiers and links.

UniProtKB protein record showing sequence, function and evidence.

Ensembl gene page showing genomic region, transcripts and exons.

Diagram of DNA → RNA → Protein.

Diagram of the bioinformatics data-science pipeline.

Example of a biological dataset represented as a matrix.

VISUAL STORY
Molecule → Dataset →
Database
→ Algorithm → Result →
Biological Meaning

Visual placeholder / suggested image

Instructor Notes — Teaching Strategy

Recommended approach for MS Data Science students

Start from a biological question, not from a software tool.

Explain biology only to the depth needed for computational analysis.

Immediately connect each concept with data structures, algorithms or statistics.

Demonstrate real databases instead of only describing them.

Ask students to interpret outputs rather than merely execute commands.

Emphasize metadata, evidence, reproducibility and validation.

Use Python/R/Bash progressively in later lectures.

TEACHING LOOP
Explain → Demonstrate →
Analyze
→ Interpret → Question

Visual placeholder / suggested
image

End of Lecture 1

Next lecture: Molecular Biology for Data Scientists

Review the key definitions.

Complete the NCBI + UniProt + Ensembl exploration activity.

Bring one biological question you would like to answer using data.

Next lecture will connect DNA, RNA, proteins, genes and the central dogma with computational data structures.

NEXT
DNA • RNA • PROTEIN
Genes • Genomes • Data

Visual placeholder / suggested
image

Sources & Further Exploration

Official resources used for the lecture content

NCBI — biomedical and genomic databases, tools and training resources.

UniProt — protein sequence and functional information.

Ensembl — integrated genome annotation and comparative genomics.

Suggested classroom practice: use current live database pages because interfaces and releases evolve.

Reference pages: ncbi.nlm.nih.gov • uniprot.org • ensembl.org

OFFICIAL RESOURCES
NCBI • UniProt •
Ensembl

Visual placeholder / suggested
image